

design being open for the maker movement to reach its full potential can never be stressed enough.

The Open Source Hardware (OSHW) movement did pick some pace around 1997, when Bruce Perens (creator of the Open Source Definition and co-founder of OSI) launched the Open Hardware Certification Programme which allowed companies to self-certify their product as Open-Source. Essentially, this reassured continued software development for the product under any circumstance.

In spite of the similar connotations, comparing the two concepts brings to light some interesting disparities. A free-program can be copied and changed without really involving the exchange of money and mostly a direct network connection between the source and the destination should suffice. So, Open Source software is commonly free in both the “free speech” sense, as well as in the “free beer” sense. But for hardware on the other hand, even if the design is freely available and distributed, the fabrication is still necessary and crucial. A design, even if digitized and transformed into a program, cannot be a substitute for hardware. And hardware, unless mass produced, involves a costly manufacturing process.

This is where the real impact of open source design shows. An open source design can be ideally made by anyone. Now, when manufacturing companies decide to join the fray, what you get is hardware whose components are known to everyone, whose cost of design is significantly lesser than that of proprietary designs. So, for an individual’s needs, free design does give rise to affordable hardware.

But affordable is still not necessarily ‘open’ to an individual’s need for modification. What would really make the hardware truly open is for individual to have the ability to fabricate their own chips according to their own requirements. The possibility of exploring the fabrication side of free hardware is not really impossible and we will come to that in a later section.

As far as the source files are concerned, it can be anything that has been used as a source to develop the product. From CAD Designs, to blueprints, sewing diagrams, anything used should be readily and easily accessible, which makes us reconsider our viewpoint regarding many OSHW offerings out there. It has been noted by many in the Maker community that many devices and products carrying the OSHW mark nowadays are not providing easily accessible documentation. Most of the times the documentation is really difficult to find out, or vague and misleading in its text. This is akin